



University of Sciences and Technology Houari Boumediene
Faculty of Physics | Laboratory of Theoretical Physics and Didactics
with the participation of The Innovative Youth Association of Algiers



SCIENTIFIC PROGRAM



FIRST NATIONAL SPACE SCIENCE WORKSHOP

October 4 – 5, 2026

USTHB, Algiers, Algeria



Sunday, October 4, 2026	
08h00 – 09h00	Welcome and registration (Faculty of Physics)
09h00 – 09h30	Official opening
09h30 – 10h30	Site visit: Solar observation Presentation of the Equipment of the Algerian Space Agency (ASAL)
10h30 – 11h00	Coffee break / Poster session (High Energy Astrophysics, Space Techniques)
CHAIRMAN: Pr. M.R. OUDIH	
11h00 – 11h50	PLENARY LECTURE ATTALLAH Reda <i>The Violent Universe in Multi-Messenger Mode: Gamma Rays, Neutrinos and Cosmic Rays</i>
CHAIRMAN: Dr. S. KALLI	
11h50 – 12h30	THEMATIC LECTURE IFTENE Meziane <i>Technological Watch and AI innovations in Space Applications: Key research activities at the Algerian Space Agency</i>
12h30 – 13h10	THEMATIC LECTURE DAMARDJI Yacine <i>Observing the Sky with a Small Telescope</i>
13h10 – 14h00	Lunch break
CHAIRMAN: Pr. R. ATTALLAH	
14h00 – 14h20	ORAL PRESENTATION BOUMECHTA Yacer <i>The Hubble tension</i>
14h20 – 14h40	ORAL PRESENTATION BOULKABOUL Amina <i>Cross-catalogue Classification of TESS Variable Stars</i>
14h40 – 15h00	ORAL PRESENTATION DJAFRI Youcef <i>Beyond Technical Obstacles: A Structural-Ontological Diagnosis</i>
15h00 – 16h00	Sunspot observation + equipment presentation
16h00 – 17h00	Coffee break / Poster session 1 (High Energy Astrophysics, Space Techniques)

Monday, October 5, 2026

CHAIRMAN: Pr. M. AKSOUH

09h00 – 09h40	PLENARY LECTURE KALLI Siham <i>Black Holes & Active Galactic Nuclei: An X-Ray Perspective</i>	PARALLEL SESSION Astrophotography Training 09h00 – 11h00
09h40 – 10h20	THEMATIC LECTURE BRADAI Djamel <i>Muonic Tomography: Considerations on a Fascinating Technique</i>	
10h20 – 11h00	THEMATIC LECTURE BOUYAHIAOUI Zineddine <i>Numerical Simulation of Meteoroid Atmospheric Entry</i>	
11h00 – 11h40	Coffee break / Poster session 2 (Astronomy and Astrophysics, Space Weather, Solar System, Cosmology)	

CHAIRMAN: Dr. A. BELHOUT

11h40 – 12h10	THEMATIC LECTURE AGUECHARI Nassim <i>Fundamentals of Space Systems: Satellite Technology, Mission Design, Remote Sensing and Satellite Subsystems</i>	PARALLEL SESSION Astrophotography Training 11h40 – 13h10
12h10 – 12h40	THEMATIC LECTURE BERABAH Abdelghani <i>High-Frequency Electrostatic Waves in Ultra-Relativistic Primordial Pair Plasmas</i>	
12h40 – 13h10	THEMATIC LECTURE BENSAID Hakim <i>From the Big Bang to the Stars: The Role of the Quark-Gluon Plasma</i>	
13h10 – 14h00	Lunch break	

CHAIRMAN: Dr. Z. BOUYAHIAOUI

14h00 – 14h30	THEMATIC LECTURE MOUSSA Djamel <i>Le paradoxe de l'âge du soleil</i>	PARALLEL SESSION Astrophotography Training
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14h30 – 14h50	ORAL PRESENTATION ZAATRI-CHABOU Amel <i>Variations in Solar Differential Rotation During Solar Cycles 23-25</i>	14h00 – 15h00
14h50 – 15h10	ORAL PRESENTATION BOUMAZA Hamza <i>Hairy Compact Objects in DHOST Theories</i>	
15h10 – 16h00	Coffee break	
16h00 – 16h30	Official closing	

Poster Session 1: High Energy Astrophysics, Space Techniques

N°	Participant	Title
P10	Souad BELLAGOUNE	X-ray Spectral Analysis of an AGN: Reflection and Variability in Mrk 841
P11	Embarka BOUTALBI	mKdV Solitary Waves and Double-Layer Structures in Quark-Gluon Plasmas: A Parametric Study
P12	Rokaya DJIDA	Analysis of the Probability Density for the Temperature-Driven QCD Phase Transition
P14	Mohamed Aris KELLOU	General Relativity within the vierbein Formalism in Astrophysics
P15	Asma Tahar TAIBA	Application of the Generalized Uncertainty Principle to Black Hole Remnants.
P16	Saliha BESKRI	The Energy Spectrum of The Hydrogen Atom in The Presence of a Minimal Length
P17	Meriem HAINE	Calculation of Charge Exchange Cross Sections between Solar Wind Protons and Neutral Hydrogen in the Martian Exosphere over the 0–2 keV Energy Range
P18	Meriem HAINE	Calculation of Radiative Excitation Cross Sections Induced by Energetic Neutral Atoms in the Martian Atmosphere
P19	Farida SAAD	Spectral Engineering for Space-Grade Photovoltaics: A DFT+Ud+Up Study of Lanthanide-Doped Rutile TiO ₂
P20	Houria CHERIEF	Optimization of Multilayer Shielding for Galactic Cosmic Ray Cancer Risk Mitigation in Long-Duration Deep-Space Missions
P21	Ibrahim CHERIFI	NIRISS Instrument of the JWST: an overview
P23	Nabila TAIBOUNI	Monte Carlo Optimization of a Backscattering Device for Planetary Subsurface Exploration

Poster Session2: Astronomy and Astrophysics, Space Weather, Solar System, Cosmology

N°	Participant	Title
P1	Imane TRABELSI	Stellar s Process Nucleosynthesis: (n, γ) Reaction in the keV Range
P2	Rayane DJOUADI	Photometric modeling of a Short-Period Eclipsing Binary TIC 273854760
P3	Maria MANSOUR	Evolution of the Sun's Internal Structure: Radial Profiles Across Key Evolutionary Stages
P4	Syham LIAS	MCTDH Wave-Packet Study of Proton–Hydrogen Charge Transfer Dynamics for Energetic Neutral Atom Production in the Heliosphere
P5	Rahma Malak BELAOUED	Dark Matter and Dark energy
P6	Messaouda BOUALI	Constraining the Generalized Uncertainty Principle via Gravitational Light Deflection
P7	Ikram BAKEL	Investigating the worlds beyond the dust: The case study of Stephenson 2 - 18
P8	Yasmina BOUDERBA	A Multi-Index Analysis of Geomagnetic Storms in Solar Cycle 24
P24	Maria MANSOUR	The Instability of Kelvin-Helmholtz in the Astrophysical Jets
P25	Sana BOUZIDI	Analysis of CME -Induced perturbations in the ionospheric D-Region using VLF remote sensing .
P26	Nour Corane MEHENAOU	Europa and Enceladus: Could They Be More Habitable Than Mars?
P27	Djihane BENMORSLI	Caractérisation du couplage atmosphère–ionosphère par une approche multi-instrument
P30	Intissar GASMI	The Impact of Friedmann Equations on Cosmic Expansion: A Workshop